what is slopware, how do you spot it, what do you do?

a curriculum that teaches you to measure the distance between something that looks like it works and something that actually holds up. every chapter ends with a check you run on your own project.

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your files?

03 why does your link only open for you?

04 what is the difference between npm run dev and npm run build?

05 why does the till stand in a different room?

00

why does this book exist?

Why this book exists, who is writing it, and how to read it.

I have been using forums such as Reddit, Quora and Ekşi Sözlük for years, and once I was one of the people posting there. People make fun of localhost:3000 users, and they believe CS is dead because somebody developed a local website. Actually, I hate needless jokes made just to fit in. But also, lack of technical depth gave those localhost:3000 owners the courage of a superhero. So instead of joining in, I decided to create a blog about how to avoid slopware, based on my own experiences.

Learning it cost me 53 repos, 8 pull requests and 6 Claude Max x20 accounts. It also cost me exploding token bills, sleepless nights, and the 5 kilos I lost to hunger and lack of sleep. I do not want that to happen to anybody else, and I do not want a power this large to stay hidden inside one class of people. I am using Claude while I write this, and I am sure Dario Amodei would want the same. So I am writing it with my own hands, out of everything I pulled from my own pain.

I do not believe that CS

has been replaced by AI.* Over time there was COBOL and C++ did not replace it, Java did not replace C++, and Python did not replace Java. Given the nature of high level languages, I treat an LLM as one more high level language. (And where has engineering gone? Engineering is the difference in the thinking itself, in knowing what to make and in connecting technologies and developing them.)

I love vibecoding

Tony Stark is a vibecoder with unlimited tokens. As far as I see, internships and projects are presented as though they belong to certain people. What gets inherited is not only money. You can call it inherited vision, you can call it social climbing, and there is even a niche for it now, something like a class based networking hub founder. What will happen soon is that somebody starts a business with a Claude Pro account and has graduates of good schools working underneath them, when those same students could have started that company with their pocket money. Nevertheless, if it is not rocket science and it is not tank design, I do not think any work that needs no large capital is as big and as irreplaceable as it is claimed to be. Your application was rejected, so say that you could do this too and carry on your way.

Everyone should benefit from technology, and not only for business and money, but for making products for your own needs. As I believe from Plato, no one does wrong willingly, and nobody writes bad software on purpose either. Outside academic purposes, CS is no different from being able to sew a button on. Everyone should learn it in order to make the ideas they dream of real, and most importantly, everyone can, because I believe that for once in history anyone can build anything.

What matters is whether anything stands behind the code. Think about the last time your terminal told you every test passed. It told you that some tests ran and none of them complained, and nothing about whether they would have noticed a broken app.

Who am I?

I am Damla, a CS student at Bilkent. I was on the board of IEEE, I worked as a Python assistant, I did an internship and a fellowship, and the rest is on my CV.

Actuall y I was a coder before CS took over, starting with HTML at 11 and a lot of Stack Overflow. Now I vibecode anyway. (Okay, what I really do is LLM systems engineerin g, which means I connect systems and engineer them.)

ir-globe draws what 198 countries do to each other and it has been going for months without me touching it, on infrastruct ure that costs 0 dollars a month. stitchu takes a photograp h of a garment and gives back a sewing pattern, and it has a gate that decides whether that pattern can actually be sewn, which I wrote after printing patterns that passed every test I had and could not be sewn at all. lulumelon measures what language models say about a brand and prints a range rather than a single number, because on its first live run the honest interval ran from 0 to 33,3. rabadon is a gate system for agents, written in C++ with no dependencies, and it exists because an agent once told me it had finished a job it had not finished.

Each of those started as something I got wrong, then found, then fixed, and the fix is what this book hands over.

How should you read this book?

The order is semantic, and I spent a full week reading and thinking before I settled it. I started from localhost, because that is the joke I kept seeing. Writing it made me realise the link only fails if you believe a page is a place, so that subject had to come first. Writing that one made me realise I was talking about files without ever saying what a folder is, so the terminal came next.

Localhost then required npm run dev, because once you know whose machine answers you want to know what is running on it. That one required npm run build, because the thing you would put on another machine is not the thing running on yours. After that came the backend, because a program on another machine is what the whole book has been walking towards. The backend required data, because a backend keeps something. Data required .env files and API keys, because the moment something is kept, whatever reaches it matters.

Those chapters are grouped into five parts. 101 is what you are actually holding. 201 gets your project off your own computer. 301 is where real users and real attackers arrive, 401 is the part people touch, and 501 is money. Afterwards I will write about LLM system design, orchestration, loops, workflows and gates.

Every chapter opens with a real incident and closes with the thing you can do tomorrow, on your own project. I don't want to stretch the schedule because 2 months in tech = 10 years of development in any topic.

By the way, new tabs keep opening in my head, and a chapter about one thing wakes up an idea about something else entirely. Those ideas do not belong inside the chapter, so they go in the appendix, and terms and small technical things that do not fit there go in the the footnotes. Both sit at the bottom of the page, so anything marked with a star has a note waiting for it.

The appendix is also where the writing about product manageme nt and the startup world will go. Companies still run events to hand out the kind of knowledge you can only get through experience , and then they turn creative people away with reasons on the level of you are a Gemini, that is why you were eliminated . To tell you the truth, I do not think anybody is as well intentioned as I am, and people hold a strange principle about not sharing good things. You can steal my ideas. There is sand in the sea and there are ideas in me, so passing on what I know is nothing but a happiness for me.

Thanks

Years ago the yazbel documentation made me interested in computer science, and I changed my departmen t. Even though we have not been writing code by hand for the last year, it gave me a love of CS that no undergrad uate degree could have given me. We used to shorten loops with algorithm analysis, and now we do the same thing inside workflows. The language changed and the tool changed, while the purpose and the logic did not, so I still think this motion in CS is a matter of perspective.

So when I am no longer a sweet student but a businesswoman, I hope this book will have enlightened somebody, entertained somebody, or made somebody think.

Two heads are better than one, so whatever work you are in, try your own idea too. My endless respect to everyone who is dreaming and working for something.

* Is CS dead? The claim is usually made about a language, not about the field. COBOL was going to be replaced by C++, C++ by Java, Java by Python, and every one of them is still running

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01

what is the thing on your screen?

A page is not a place. It is a delivery that happens again every time somebody looks.

The first thing you do with a coding agent is ask it for a website, and a minute later there is one. The terminal prints an address. You open it and the thing you described is sitting in the browser looking finished.

Ask where that thing lives and the answer comes back as in the browser or on my computer or on the internet. All three feel fine right up until something breaks, and then none of them tell you where to look.[1]

---

1 **A book note.** A model takes the easiest road. It gave you localhost with no backend, because that is the shortest

way to something that runs.

What arrives when a page opens?

Make a folder anywhere you like and put a single file in it called index.html with one line inside.

```
<h1>hell
```

Open a terminal in that folder and run python3 -m http.server 8000. Go to localhost:8000 in your browser. The word hello is there in large letters, arriving from a server small enough that nothing can hide inside it.

Now right click on the page and choose view source. What comes up is the line you typed, character for character.

Change the h1 to a p and refresh. The same word comes back small. Nobody sent you a smaller word. You changed one instruction and the browser drew the letters differently, because what travelled to it was never a page. It was a set of instructions for a program that carries them out.

How does the browser learn about the second file?

One file is not how anything real arrives. Next to index.html make a file called style.css containing h1 { color: crimson; } and mention it from the page.

```
<link re
<h1>hell
```

Refresh and the word is red.

Your browser had no way of knowing that style.css existed. The only file it had been given was index.html . It read that file and came to a line mentioning a stylesheet. Only then did it go back to the server for a second file. The page tells the browser what else to fetch, one discovery at a time.

Open the developer tools with Cmd and Option and I on a Mac or

F12 on Windows. Click the Network tab and refresh with the panel open. Two rows come up, index.html first and style.css second, and they can never come in the other order.

A slow page is rarely one slow file. The document arrives and mentions something. That something arrives and mentions two more. Nothing further down the chain starts until the thing before it has been read.

Where does the page go when the server stops?

'The page is on my computer, so it will still be there,' you may be thinking. In the Network panel there is a checkbox marked Disable cache. Tick it so the browser stops keeping copies of its own. Now go to the terminal and stop the server with Ctrl and C, then refresh the page.

The page is gone and there is no error about your HTML, because your HTML never arrived. Nothing had been sitting in the browser waiting for you.

Start the server again and refresh. Hello comes back in red exactly as before, drawn from nothing. A page is a delivery that happens again every time somebody looks.

What does this look like

in a real project?

Open the project you have been working on and refresh it with the panel open. The same list appears, only far longer. The document sits at the top. Everything the document mentioned sits under it, and under those sits everything they mentioned in turn.

At the bottom of the panel there is a total size. Every visitor downloads that much before they see anything at all, out of their connection and their phone plan.

That leaves one question for the rest of the book. Some program on some machine sent those files. On the hello page you know which one, because you started it yourself and it is still sitting in your terminal. On your real project you have probably never thought about it.

CHECK

Open the

Network panel on the project you are proudest of and refresh it.

How many files arrived?

How many kilobyte

s did they come to?

Which file in that list can you not account for? There is nearly always one. Usually a font yo

> u stopped using months ago or a library that arrived attached to something else. Nothing announced it, because it

> was mentioned by a file that was itself mentioned by a file.
>
> Nothing needs removing today. It has been going

out to every visitor since the day it appeared.

- [ ] `my project passes this check`

You just started a server inside a folder without my having said what a folder is, and that gap has to close before we can ask whose machine your files come from. So

the
terminal
and the
folder
come next.

02

why can your depl oy not find your files?

The same command run one level up serves nothing. Most deploy failures start here.

You ran a command inside a folder a moment ago and a server appeared, and I never said which folder it was reading or why that mattered. The same command run from one step higher up would have served nothing.

A whole family of failures comes out of that. The build that works on your machine and returns file not found on the deploy is one, the image that loads locally and 404s in production is another. In every one of them a program is standing somewhere you did not expect, reading a path that means something different from there.

Where are you standing?

Open a terminal. On a Mac it is called Terminal. On Windows, use the one your development tools installed rather than the old black box.

A command never runs in general. It runs while you are standing somewhere and where you are standing changes what it does.

```
pwd
```

Print working directory. It answers with the absolute path of where you are standing right now.

```
ls -F
```

Plain ls lists files and folders together with no way to tell them apart, and -F puts a slash on the end of every folder name. Do not try to work it out from the name. README has no dot and is a file, while a folder called backup.old has one and is still a folder.

```
cd Deskt
```

Change directory, into the Desktop that sits inside wherever you already are. Run pwd again and the answer has grown by one step. Run ls -F and the contents have changed, because the same command means something different here.

```
cd ..
```

Two dots is a real entry that exists inside every folder on the disk and it points at that folder's parent.

That is the whole of navigation. Three commands and one convention, and they work the same way on every Mac and every Linux machine and inside the terminal your Windows tools installed.

Is the terminal a different place?

If the terminal still feels like a second machine hiding underneath the one you use, open

Finder or Explorer and go to your project folder. Put that window on one side of your screen and the terminal on the other. Now cd your way to the same folder and run ls -F.

They are the same thing. The icons on the left and the names on the right are one folder described in two ways, and the three commands you just learned will reach any folder on the disk.

What is a path?

A folder holds files and other folders and those hold more. All of it hangs down from one starting point, so your whole disk is a single tree.

A path is directions through that tree and pwd printed one of them.

```
/Users/
```

It means start at the top of the disk and go into Users, then into damla and Desktop and project. Each slash is a door you walk through.

A postal address is written the same way and a path is that written on one line.

There are two kinds.

```
/Users/
style.c
```

The absolute one begins with a slash and is measured from the top of the disk, so it means the same file wherever you use it. The relative one is measured from where you are standing, so those same characters point at different files depending on where they are read.

Why is this so hard to learn?

Go back to the folder with index.html and style.css in it and start the server again. The page loads and the word is red. Now stop the server, run cd .. to step up one level, and start it again from there.

```
cd ..
python3
```

Open localhost:8000. There is no page, only a list of folder names.

Nothing was moved and nothing was deleted. The href inside your HTML still says style.css and it still means the style.css next to me, and the server is now standing one level up where no such file sits. Your whole project is intact and unreachable.

That is the entire failure, and here is why it stays invisible until it costs you a day. A few years ago lecturers in physics and engineering started running into something they could not explain. They would ask a class to open a file from a particular directory. We as students usually struggle to do exactly that, myself included, even though we use a computer all day and use it well.

A file lives in the app or on the computer, and the way you reach it is to search for it.

That model works for daily life. Put everything in one pile and search when you need something. You will find your holiday photographs every time. It stops working at the deploy, because the machine you deploy to has no search box. It is handed a path and goes exactly there. If nothing is there it stops.

Directory structure is so ordinary to the people who know it that words for it are hard to find, so nobody explains it and nobody asks.

Why does this decide whether things work?

Every build step, every deploy tool and every error message you will read for the rest of your life assumes you know where you are standing.

File does not exist almost always means a relative path was read from a folder you were not expecting. The file was there all along, and the program was standing somewhere else when it went looking for it. You saw exactly that a minute ago with a server one level too high.

Something works on your machine and dies on the deploy for the same reason. You typed the command while standing inside the project folder. The other machine runs it while standing wherever it happens to start, so the relative path points at a place that does not exist. Chapter 16 is where this bites hardest.

MIT teaches a free course for exactly this gap called The Missing Semester of Your CS Education. Julia Evans draws comics about the shell that are the friendliest thing on the subject.

> CHECK
>
> Open a terminal and leave your editor

> closed.
>
> Get to your project folder using only pwd, ls -F and cd. Read what comes back after each step ins

> tead of typing the whole path from memory.
>
> When you arrive, which folder in that list can you not describe the conten

> ts of? There will be one, usually node_modules or dist or .next.
>
> Go into it and look, changing nothing. The ne

> xt time something breaks that folder is going to appear in the error message (which should not be the first time yo

> u see the na me).
>
> ---
>
> - [ ] `my project passes this check`

Next comes localhost, and the reason a link that opens instantly for you does not open for your friend.

03

why does your link only open for you?

Your machine answers its own name. Your friend's machine answers its own, and finds nothing.

This chapter is about famous localhost:3000. Jokes about it have been going around the internet for a long time. Many people believe CS is dead because they built a website hosting on local. You click your own link and it opens. Your friend clicks it and it does not. Why does it happen, what is localhost:3000, and how to fix this issue?

What does localhost mean?

It begins with localhost:3000 or 127.0.0.1:3000, which are the same address written two ways. localhost is a name and 127.0.0.1

is the number the name stands for. RFC 6761 reserves both for one purpose, and that purpose is to mean the computer that is asking. When you type it, your machine sends the request to a program on itself. The request never leaves the machine, which is why this is called the loopback address. When your friend types it, their machine sends the request to a program on itself and finds none.

Each computer's localhost is a different server.

The number after the colon is the port. If the address is the building, the port is the apartment inside it. One machine runs many programs and each waits at its own number. 3000 is a convention developme nt tools picked, nothing more. The port is fine. The building is the problem, because you gave your friend the name every building uses for itself.

Can your own phone open it?

Open the terminal and cd into any folder that has a file or two in it. Run this.

```
python3
```

This starts a small web server. The flag tells it to answer only the machine it is running on. Open localhost:8000 in the browser and you see the folder's files listed. Now type the same address into your phone's browser. Nothing comes back, even though the phone is on the same wifi. The phone went looking for a program at port 8000 on itself.

Now stop the server with Ctrl+C and start it again without the flag.

```
python3
```

The flag was doing all the work. 127.0.0.1 told the server to listen on the loopback address alone, so requests arriving over wifi were never answered.

Without the flag this tool listens on 0.0.0.0, which is not an address you connect to. It means every network interface the machine has, including the wifi card. A request from the phone is now accepted the same as one from the machine itself. You will see 0.0.0.0 again in Docker logs and in cloud deploy output and it always means the same thing.

Now find the laptop's address on the local network. On a Mac that is ipconfig getifaddr en0 and on Linux hostname -I. It looks like 192.168.1.24. If the Mac command comes back empty your wifi is on another interface, so try en1. Type http://192.168.1.24:8000 into the phone and the file list appears. Nothing in the files changed, only which doors the program answers and which machine the phone was told to look at.

How far does the local network reach?

'So I drop the flag and I am live,' you may be thinking. What you have is a server reachable from the same wifi. 192.168.1.24 is an address inside your home network and means nothing outside it, so a friend in a cafe or a customer on mobile data cannot reach it. And when the laptop lid closes, nothing is running anywhere.

A product has to answer while you are asleep, so the program has to run on a machine that stays on and has an address the internet can route to. Putting it there is called deploying, and that is chapter 16. ir-globe answers at three in the morning because it is not on my laptop.

> CHECK
>
> Look at the

> address bar of the project you are proudest of. If it begins with localhost or 127.0.0.1, the number of people wh

> o can open it is 1. Then close the laptop. Turn off wifi on your phone so you are on mobile data and try to op

> en the project. Whatever comes back is the answer to whether it exists yet.
>
> ---
>
> - [ ] `my project passes this check`

After that come the two commands you run without reading them. Only one of npm run dev and npm run build makes a thing you can deploy.

04

what is the differ ence betw een npm run dev and npm run build ?

One builds pages in memory for you. The other writes the folder your users get.

Localhost turned out to be a name your machine uses for itself. The other half of the address is still unexplained, because something is answering at that door, and in a framework project that something is npm run dev.

You have run that command a thousand times without wondering what it is. Sitting next to it in the same file there is npm run build. It takes much longer and produces a folder you have probably never opened. That folder is the part your users get.

What is running when you run npm run dev?

What runs is a program sitting in your terminal and waiting to be asked for things, the same way python3 -m http.server did in chapter 1. This one is far cleverer about what it hands over.

Go to the terminal

where it says ready and press Ctrl and C. Refresh the browser. The site is gone and the error is a refused connection, which means nothing is listening at that door any more.

Now start it again and open the project folder, looking for the HTML that arrived in the browser. There is no such file. Your project has a src folder full of components and not one of them is a page. Nothing on the disk matches what you saw in view source.

The dev server built that page in memory the moment your browser asked for it and then let it go. Its whole job is to read the source, work out what the browser needs right now and hand it over. Then it watches the files so it can do the same thing the instant you save. Nothing was written down, so a save appears in the browser in under a second.

What does the build do?

```
npm run
```

It takes a while, where the dev server started instantly. When it finishes there is a new folder called dist on Vite and .next on Next.js. Go in there with the commands from chapter 2.

```
cd dist
ls -F
```

The HTML that did not exist a minute ago is in there. So are the scripts, with names like index-4f3a9c.js. Those characters in the middle are there so a browser can safely keep an old copy without mistaking it for the new one after you deploy. The CSS from every component in the project has been gathered into one file.

Your source is written in whatever shape lets you work. This folder is written for browsers, in one shape and once. The dev server had been standing in for it on demand without ever writing a word of it to disk.

Which one gets deployed?

The folder gets deployed, never your source and never the dev server. Deploying means building that folder and putting its contents somewhere that answers requests all day, so a visitor is handed a finished file instead of waiting for a program to work one out.

You can serve that folder

yourself right now.

```
cd dist
python3
```

Open localhost:8000. That is your site, coming out of a plain file server that has never heard of your framework. It is the same server that handed you one line of HTML in chapter 1.

Why do the two behave differently?

'Same code, same project, so it will behave the same way,' you may be thinking.

Open one of those scripts in dist and read it. Your function names are gone, replaced by single letters. The spaces are gone. The whole file is one line.

The dev server is optimised for you. It starts instantly and refreshes instantly, keeps your variable names so errors stay readable, and squashes nothing. The build is optimised for the visitor, so it makes the smallest files it can, squashes everything together, drops code that nothing uses and shortens every name it can reach.

Squashing changes behaviour. Code that quietly relied on a function still having its own name breaks, because the name is gone and you just read the file where it went missing. A value read while the build is running gets frozen into the file, so changing it on the server afterwards changes nothing until you build again, and chapter 13 is about the trouble that causes. A file the dev server found by walking your folder is missing from the output entirely if nothing imported it properly.

So a project can run beautifully for weeks and fall over on the first deploy without a line of source having changed. When somebody says it works on my machine they are usually telling the truth, because their machine was running the friendly version.

Before you deploy, run the build and serve the output locally the way you just did, then click around your own site. Anything broken in that folder is broken in production, and you get to find it while you are sitting in front of it.

CHECK

Run npm run build on your

own project and let it finish.

How long did it take? Whatever that number is, it happens again on every deploy.

What is inside the folder? Go in with cd dist or cd .next and run ls -F, then find the HTML your source never contained.

Then serve it with python3 -m http.server 8000 and use your own site coming out of a file server

> that knows nothing about your framework. If something works under npm run dev and not in there, you have

> found a real bug months before a stranger would have found it for you.
>
> ---
>
> - [ ] `my project passes this check`

Chapter 5 goes to the shop front and the till. What a backend is, and why some things can never live in the files you just looked at however well you hide them.

05

why does the till stand in a different room?

Everything delivered to the browser belongs to whoever is standing there.

Everything so far has happened in one room. Files arrive and the browser draws them. You have now read your own project the way a visitor reads it. A shop has a second room behind the counter where the till and the keys are, and nobody has to be told why. In a project both rooms look like folders sitting next to each other in the same editor, so the till ends up out on the shop floor.

Which room is

the browser?

The browser is the front one and everything in it belongs to whoever is standing there.

Every file that arrived is listed in the Network panel and you can click any of them and read it, including the scripts you wrote. That will not be patched one day, because it is what delivery means.

'My code is minified, so nobody can read it,' you may be thinking. Open one of your own scripts in that panel and find the button marked with braces at the bottom of the view. The one line becomes readable code again. Squashing a script makes the file smaller and it hides nothing.

What is a backend?

A backend is a program on another machine that answers requests. It runs somewhere else and listens at a door. You talk to it by sending exactly the kind of request you watched arrive in chapter 1.

The customer orders and the kitchen cooks. The customer does not come into the kitchen, because the kitchen is a different room. A script running in the browser is a locked cupboard standing in the dining room, and the customer has all evening.

What is an API?

An API is the menu. A menu tells you

what you may order and in what form. An API tells you which requests the kitchen accepts, meaning this address with this method and these fields. Anything not on the menu is refused. Writing one is the act of deciding what strangers may ask you for.

The verbs on that menu are the methods. GET fetches something, POST sends something new, PUT replaces, DELETE removes. A GET is understood everywhere to only fetch and search crawlers act on that understanding. That is how a link which deletes a row gets triggered by a crawler that was only looking around.

What do the numbers mean?

Every answer comes back with a status number and MDN carries the definitions if you want them properly.

200 means it worked. 404 means

the kitchen understood you perfectly and has no such dish. 500 means the kitchen broke while cooking, so the fault is on their side.

The two that get confused constantly are 401 and 403. A 401 means the kitchen does not know who you are, while a 403 means it knows exactly who you are and refuses anyway. Identity and permission are separate systems with separate fixes, and telling them apart is the whole of chapter 26.

Can you break your own rule?

Find something in your own project that the page refuses to let you do. A submit button that stays disabled until the form is valid, a field with a maximum length, a section that only appears when you are logged in. Anything the page decides.

Open the developer

tools and go to the Console tab. Then type one of these, matching whichever rule you found.

```
document
document
```

The button is live. The field takes as much as you want. You did not find a hole in your code, you retyped a line of it, and every visitor has the same console you just used.

That is why a check written into your page is a suggestion. Your script runs on their machine. A check that holds has to run in a room they cannot reach, which is the room this chapter is about.

Where does the key go?

Into the kitchen and never into a request the customer can read.

stitchu sends a photograph to a vision model and that model charges per call. The key that pays for it belongs in the till, so the browser never holds it. The photograph goes to a Cloudflare Worker, the Worker holds the key and calls the model, and the answer comes back. Anyone can open the Network panel on stitchu and read every request that leaves their machine and find nothing in them to take.

The wrong version is one line of difference. The browser calls the model directly with the key attached to the request. It works on the first try (which is the trouble) because from behind the counter both versions look the same to you. Every visitor now holds a working key billed to your account, and chapter 31 is about the size of that bill.

> CHECK
>
> Open your own project with the Netw

> ork panel open and use it until requests start appearing.
>
> Where is each request going? Read the address of each

> one. If it goes straight to somebody else's service then your browser is talking to that service directly and

everything in that request is public.

What in your code costs money or grants access? An API key, a database UR

> L, an admin token. Where does it live right now?
>
> If it reached the browser then it is public, and the next tw

> o chapters are about moving it.
>
> ---
>
> - [ ] my project passes this check

Chapter 6 goes to where data lives, and why a file works perfectly until two people use your project at the same time.